

Scientific Notes: Spatial Structure Analysis of Energy Patterns

Objective: Investigate the spatial structure and dynamical characteristics of Energy Pattern 1 (EP1) and Energy Pattern 2 (EP2) cyclones during intensification phase.

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1. Introduction

1.1 Motivation

Following the identification of three distinct energy patterns in South Atlantic cyclones through PCA-based K-Means clustering (see `cluster_analysis_energy_patterns/SCIENTIFIC_NOTES.md`), this analysis investigates the **spatial structure** of the most energetically active patterns.

Energy Pattern Characteristics (from cluster analysis): - **EP1 (11.6%)**: Large barotropic and baroclinic conversions; **exports energy** to surroundings - **EP2 (25.6%)**: Moderate balanced conversions; **imports energy** from large-scale environment
- **EP3 (62.7%)**: Weak energetics representing typical “day-to-day” cyclones

Rationale for EP1 and EP2 focus:

EP3 cyclones exhibit weak energy budget activity and represent the climatological background. EP1 and EP2 cyclones, being energetically distinct and more intense, are the primary focus for understanding what structural characteristics differentiate high-impact systems from typical cyclones.

1.2 Research Questions

1. What are the spatial structures of baroclinic instability (EGR) in EP1 vs EP2?
2. How do upper-level (200 hPa) and low-level (850 hPa) PV anomalies differ between patterns?
3. What thermal advection patterns characterize each energy pattern?
4. How does near-surface moisture distribution and convergence differ?

Approach: Composite analysis of ERA5 reanalysis fields during the intensification phase of all EP1 (N=444) and EP2 (N=979) cyclones.

2. Dataset

2.1 Sample Composition

Energy Pattern	Cluster ID	Cases Analyzed	Percentage
EP1	0	444	11.6%
EP2	2	979	25.6%

Source: All EP1 and EP2 cyclones from cluster analysis with complete lifecycle phases.

Temporal Coverage: Intensification phase only (6-hourly timesteps during deepening period)

2.2 ERA5 Reanalysis Data

Variables Downloaded: - **Pressure levels (hPa):** 175, 200, 225, 250, 500, 825, 850, 875, 975
- **Pressure-level variables:** u, v, t, z, q (winds, temperature, geopotential, specific humidity) -
Single-level variables: msl (mean sea level pressure)

Spatial Configuration: - **Resolution:** $0.25^\circ \times 0.25^\circ$ - **Domain:** $30^\circ \times 30^\circ$ centered on cyclone track center - **Composite domain:** $15^\circ \times 15^\circ$ (marked in figures)

Temporal Resolution: 6-hourly

3. Methodology

3.1 Diagnostic Fields Computed

Diagnostic	Level(s)	Purpose	Key Reference
Eady Growth Rate	250–850 hPa	Baroclinic instability measure	Lindzen & Farrell (1980)
Potential Vorticity	200 hPa	Upper-level dynamics, tropopause folding	Hoskins et al. (1985)
Potential Vorticity	850 hPa	Low-level PV anomaly, diabatic effects	Čampa & Wernli (2012)
Temperature Advection	850 hPa	Warm/cold advection patterns	Sanders & Gyakum (1980)
Specific Humidity	975 hPa	Near-surface moisture distribution	-
Moisture Flux Divergence	975 hPa	Moisture convergence (convective potential)	Banacos & Schultz (2005)
Sea Level Pressure	Surface	Cyclone intensity and position	-
RK criterion (Rayleigh-Kuo)	250 hPa	Barotropic/baroclinic instability condition	Rayleigh (1880); Kuo (1949)
KE Advection	250 hPa	Kinetic energy tendency from advection	-

3.2 Spherical Grid Spacing

All horizontal derivatives account for Earth’s spherical geometry:

Meridional spacing:

$$dy = R_{\oplus} \Delta\phi$$

Zonal spacing:

$$dx = R_{\oplus} \cos(\phi) \Delta\lambda$$

where $R_{\oplus} = 6.371 \times 10^6$ m, ϕ = latitude, λ = longitude.

Implementation: `compute_spherical_grid_spacing(lat_1d, lon_1d)` - Verifies coordinate monotonicity

- Correct gradient signs regardless of coordinate direction - Uses `numpy.gradient()` for non-uniform grids

3.3 Eady Growth Rate (EGR)

Formula:

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

where: - $f = 2\Omega \sin(\phi)$ = Coriolis parameter - $N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}$ = static stability (Brunt-Väisälä frequency) - $\left| \frac{\partial \vec{V}}{\partial z} \right|$ = vertical wind shear magnitude

Layer: 250–850 hPa (captures main tropospheric baroclinic zone)

Quality Control: - $N^2 > 10^{-6} \text{ s}^{-2}$ (exclude statically unstable regions) - $|\phi| > 5^\circ$ (avoid equatorial singularity) - $\sigma_{EGR} < 5 \text{ day}^{-1}$ (cap unrealistic values)

3.4 Potential Vorticity

Formula:

$$PV = -g (\zeta_\theta + f) \frac{\partial \theta}{\partial p}$$

where ζ_θ = relative vorticity on isentropic surface, θ = potential temperature.

Levels: - **200 hPa:** Uses 175, 200, 225 hPa (upper-level tropopause dynamics) - **850 hPa:** Uses 825, 850, 875 hPa (low-level diabatic PV anomaly)

Units: $\text{K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ (SI), converted to PVU ($1 \text{ PVU} = 10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$) in figures

3.5 Temperature Advection

Formula:

$$\text{advT} = -\vec{V} \cdot \nabla T = -\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right)$$

Level: 850 hPa

Sign Convention: Positive = warm air advection; Negative = cold air advection

Units: K h^{-1} (converted from K s^{-1})

3.6 Moisture Fields at 975 hPa

Specific Humidity (q): - Direct measure of atmospheric moisture content - Units: g kg^{-1} (converted from kg kg^{-1}) - Level: 975 hPa (near-surface, above boundary layer)

Moisture Flux Divergence:

$$\nabla \cdot (q\vec{V}) = \frac{\partial(qu)}{\partial x} + \frac{\partial(qv)}{\partial y}$$

- **Negative values** $\rightarrow$ Moisture convergence (convective potential, latent heat release)
- **Positive values** $\rightarrow$ Moisture divergence (evaporation/drying)
- Units: $\text{g kg}^{-1} \text{ h}^{-1}$ (converted from $\text{kg kg}^{-1} \text{ s}^{-1}$)
- Key diabatic process in cyclone intensification (Banacos & Schultz, 2005)

3.7 Rayleigh-Kuo Stability Criterion (250 hPa)

Formula:

$$RK = \beta - \frac{\partial^2 u}{\partial y^2}$$

where: - $\beta = \frac{\partial f}{\partial y} = \frac{2\Omega \cos(\phi)}{R_\oplus} =$ Rossby parameter - $\frac{\partial^2 u}{\partial y^2} =$ meridional curvature of zonal wind

Physical Interpretation: - **Negative values** $\rightarrow$ Necessary condition for barotropic/baroclinic instability satisfied - **Combined criterion:** Incorporates planetary vorticity gradient and wind shear curvature - Level: 250 hPa (jet stream level, maximum barotropic effects)

References: Rayleigh (1880), Kuo (1949), Charney & Stern (1962)

3.8 Kinetic Energy Advection (250 hPa)

Formula:

$$\text{KE}_{\text{adv}} = -\vec{V} \cdot \nabla(KE) = -\vec{V} \cdot \nabla \left(\frac{1}{2}(u^2 + v^2) \right)$$

Physical Interpretation: - **Positive values** $\rightarrow$ KE advection causes local acceleration (energy gain) - **Negative values** $\rightarrow$ KE advection causes local deceleration (energy loss) - Level: 250 hPa (jet stream level) - Units: $\text{m}^2 \text{ s}^{-3}$

Significance:

Quantifies energy transport within the jet stream. Positive advection indicates regions where the flow pattern favors kinetic energy accumulation, potentially intensifying upper-level divergence and cyclone development.

4. Results

Note: Statistical summaries presented below are computed from spatial composites centered on cyclone genesis locations. Each subsection presents the composite figure followed by quantitative statistics. Physical interpretation of spatial patterns and EP1 vs EP2 differences is currently under analysis.

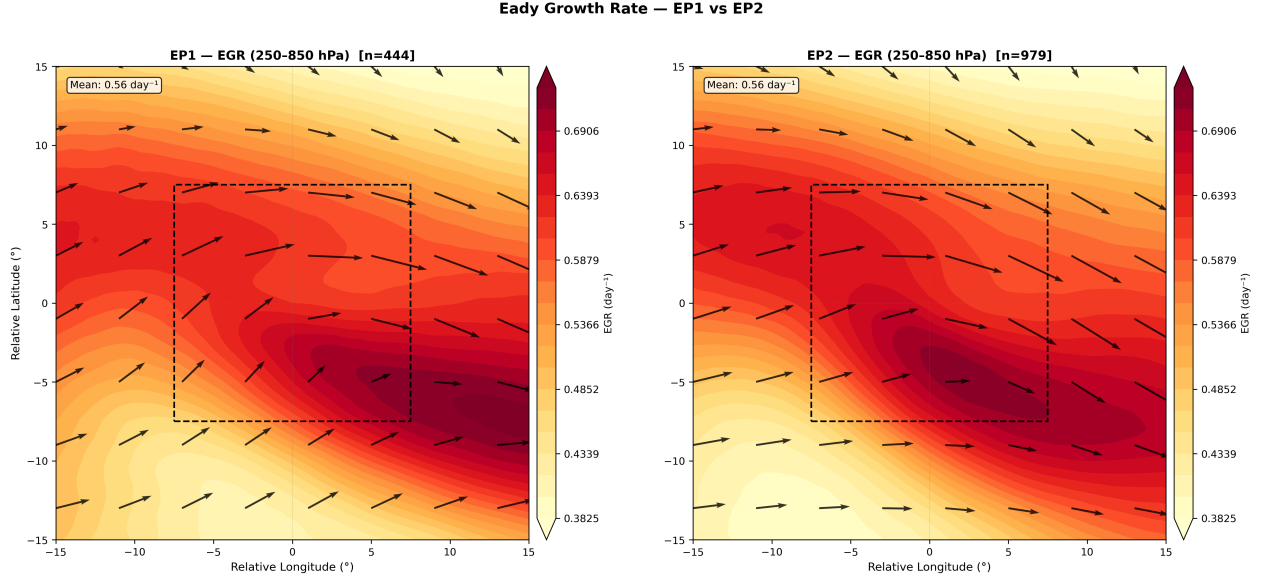


Figure 1: EGR Composite

4.1 Eady Growth Rate (Baroclinic Instability)

Figure: Eady growth rate composite (250–850 hPa layer) for EP1 (left) and EP2 (right). Contours show growth rate in day⁻¹. The 15° × 15° box marks the LEC computation domain.

Summary Statistics:

Statistic	EP1	EP2
Mean ± SD (day ⁻¹)	0.56 ± 0.09	0.56 ± 0.10
Median (day ⁻¹)	0.57	0.58
Range (day ⁻¹)	[0.37, 0.73]	[0.36, 0.72]

4.2 Upper-Level Dynamics (200 hPa PV)

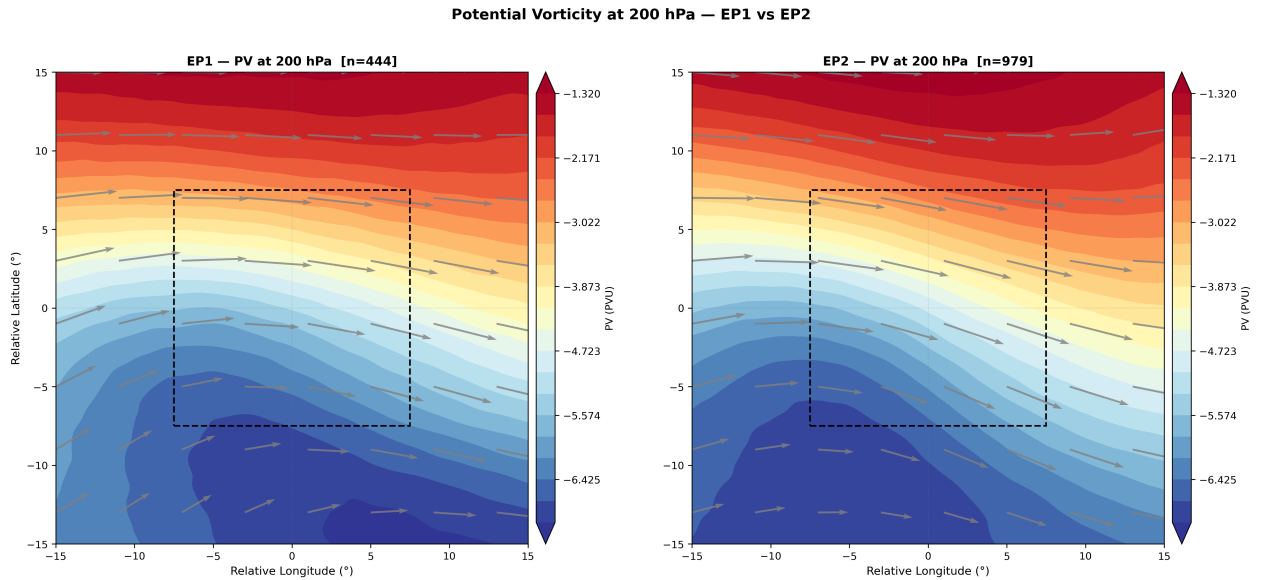


Figure 2: PV@200 Composite

Figure: Potential vorticity at 200 hPa for EP1 (left) and EP2 (right). Units: PVU ($10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$). Shows tropopause dynamics and stratospheric intrusions.

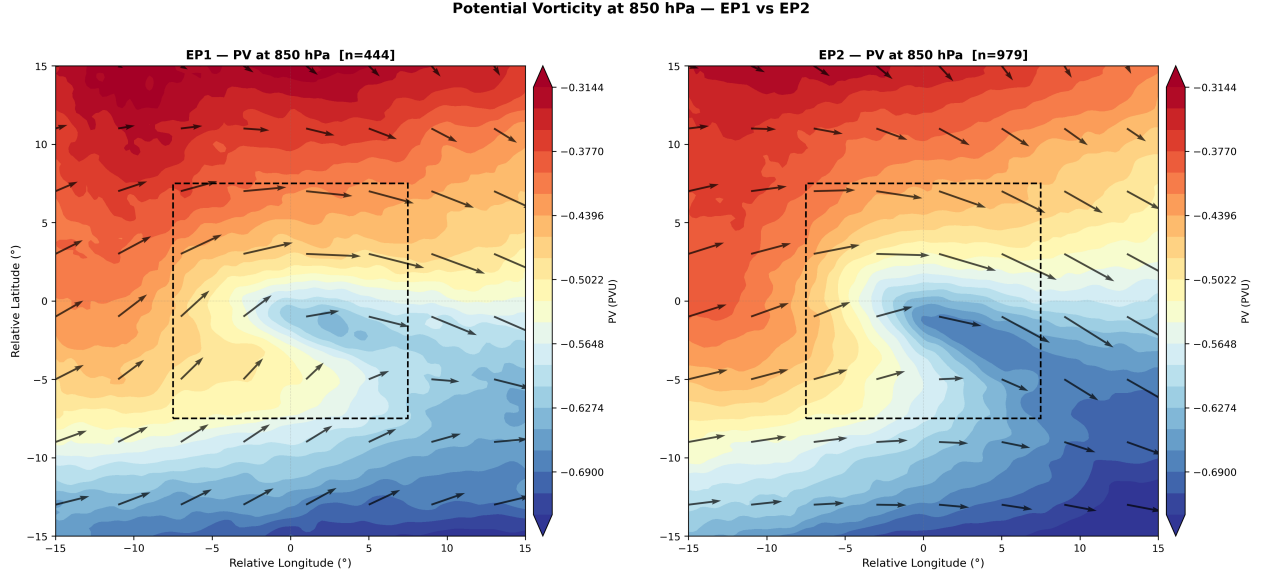


Figure 3: PV@850 Composite

Statistic	EP1	EP2
Mean (PVU)	-0.50	-0.52
Range (PVU)	[-0.74, -0.30]	[-0.75, -0.31]

4.4 Temperature Advection (850 hPa)

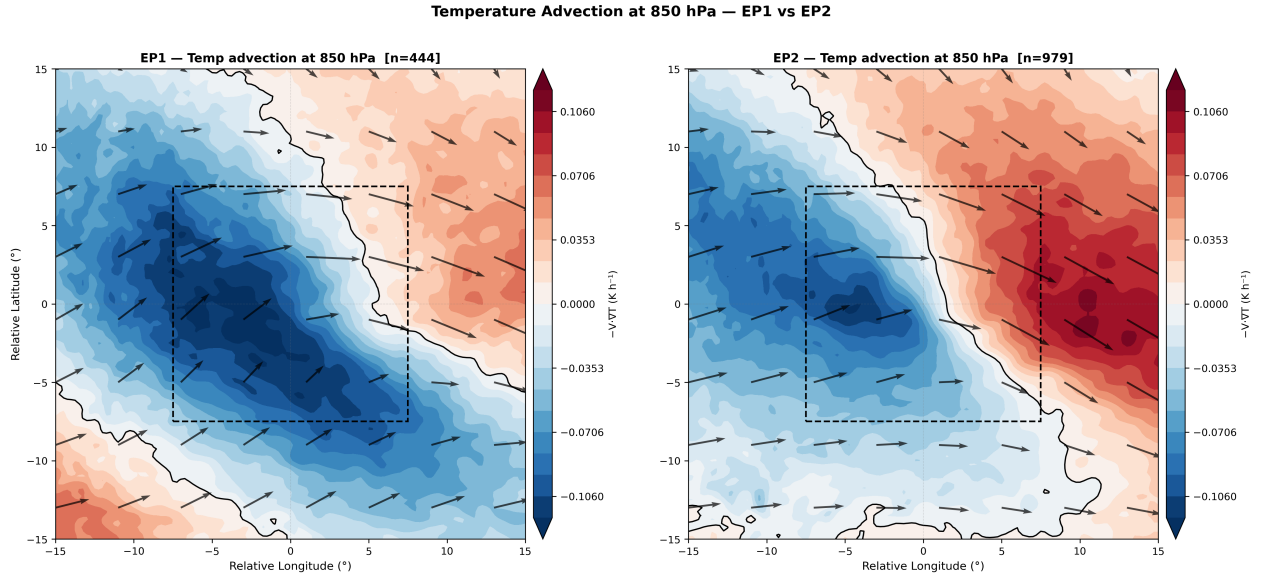


Figure 4: Temp Advection Composite

Figure: Temperature advection at 850 hPa for EP1 (left) and EP2 (right). Units: $K h^{-1}$. Positive (red) = warm advection; Negative (blue) = cold advection.

Summary Statistics:

Statistic	6	EP1	EP2
Domain mean ($K h^{-1}$)		-0.031	-0.003
Max warm advection ($K h^{-1}$)		0.071	0.111
Max cold advection ($K h^{-1}$)		-0.132	-0.119

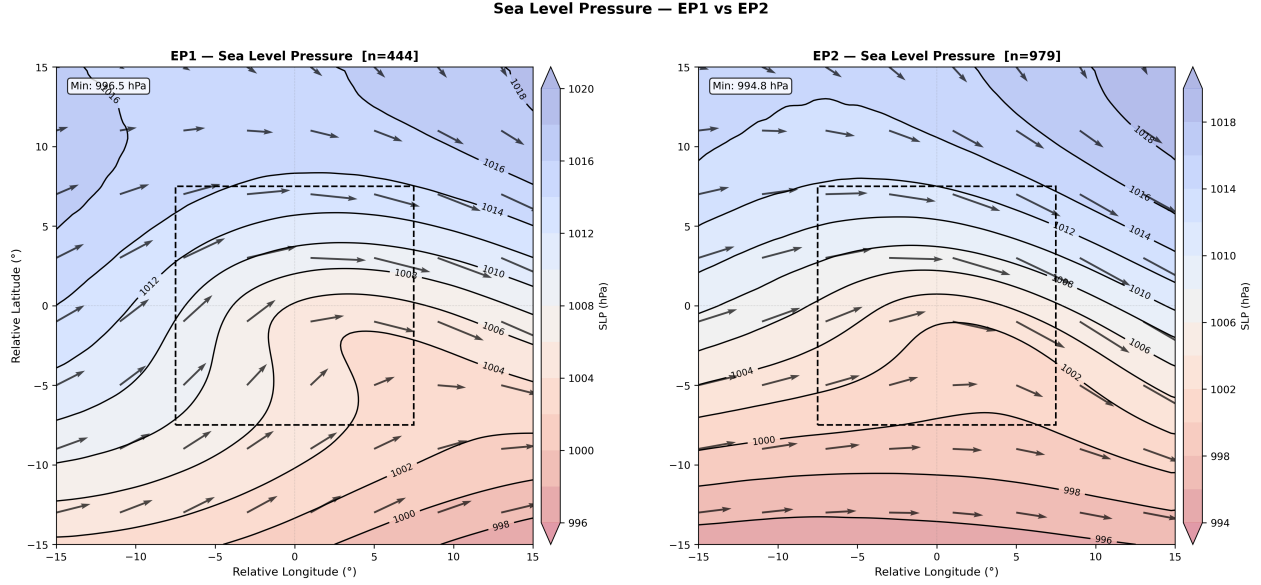


Figure 5: SLP Composite

4.5 Sea Level Pressure

Figure: Mean sea level pressure for EP1 (left) and EP2 (right). Units: hPa. Shows cyclone intensity and horizontal structure.

Summary Statistics:

Statistic	EP1	EP2
Minimum (hPa)	996.5	994.8
Maximum (hPa)	1018.3	1019.2
LEC $15 \times 15^\circ$ mean (hPa)	1008.1	1005.2
Full $30 \times 30^\circ$ mean (hPa)	1009.3	1006.4

4.6 Specific Humidity (975 hPa)

Figure: Specific humidity (shading, g kg^{-1}) and moisture flux vectors at 975 hPa for EP1 (left) and EP2 (right).

Summary Statistics:

Statistic	EP1	EP2
Mean (g kg^{-1})	6.37	6.52
Range (g kg^{-1})	[2.79, 11.14]	[3.13, 11.16]
LEC $15 \times 15^\circ$ mean (g kg^{-1})	6.31	6.58
Full $30 \times 30^\circ$ mean (g kg^{-1})	6.37	6.52

4.7 Rayleigh-Kuo Stability Criterion (250 hPa)

Figure: Rayleigh-Kuo criterion at 250 hPa for EP1 (left) and EP2 (right). Units: s^{-1} . Negative values (blue) indicate regions satisfying the necessary condition for barotropic/baroclinic instability. Wind vectors show 250 hPa flow.

Summary Statistics:

Statistic	EP1	EP2
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Low-Level Moisture Transport at 975 hPa — EP1 vs EP2

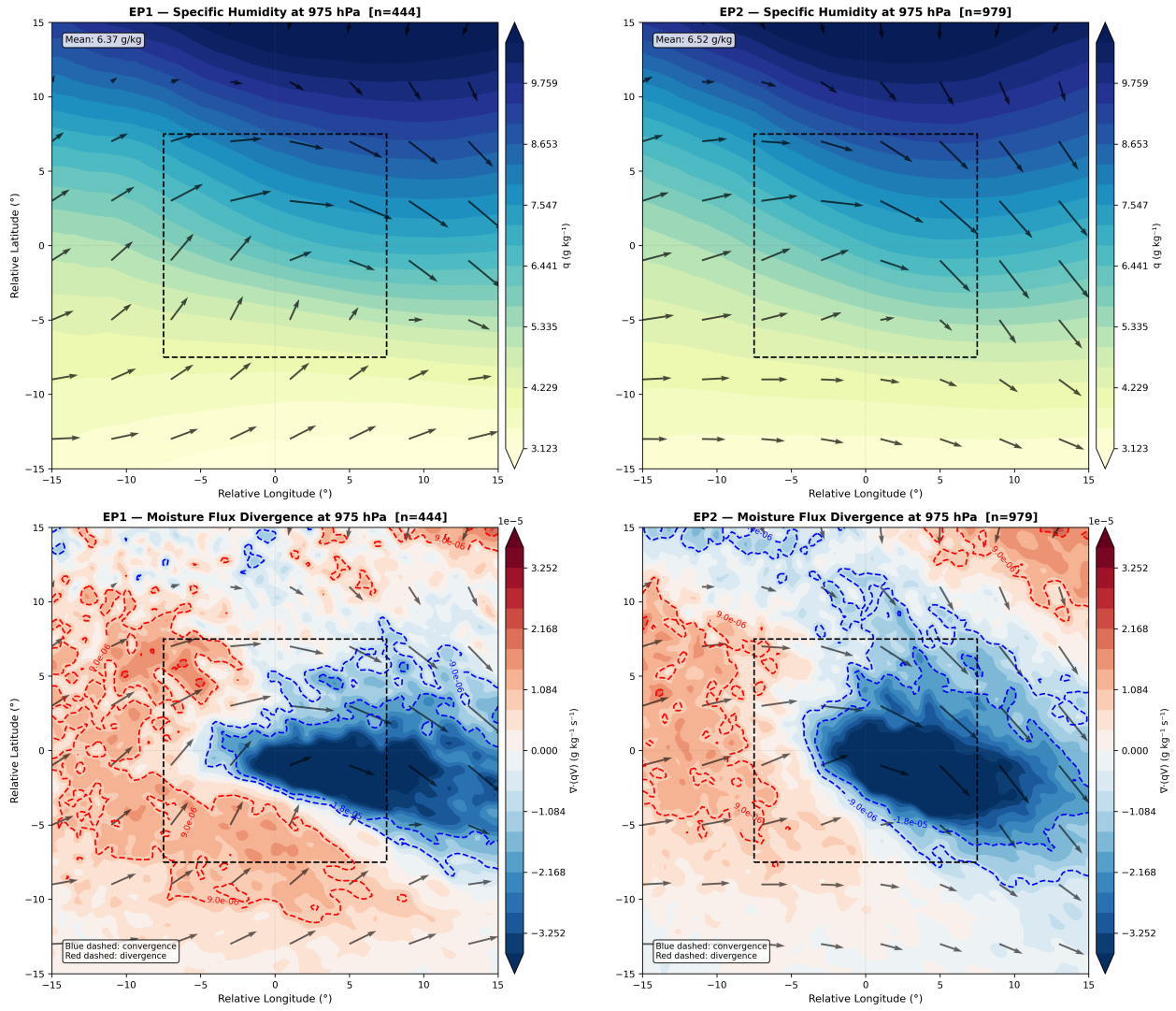


Figure 6: Moisture Composite

Rayleigh-Kuo Instability Criterion at 250 hPa — EP1 vs EP2

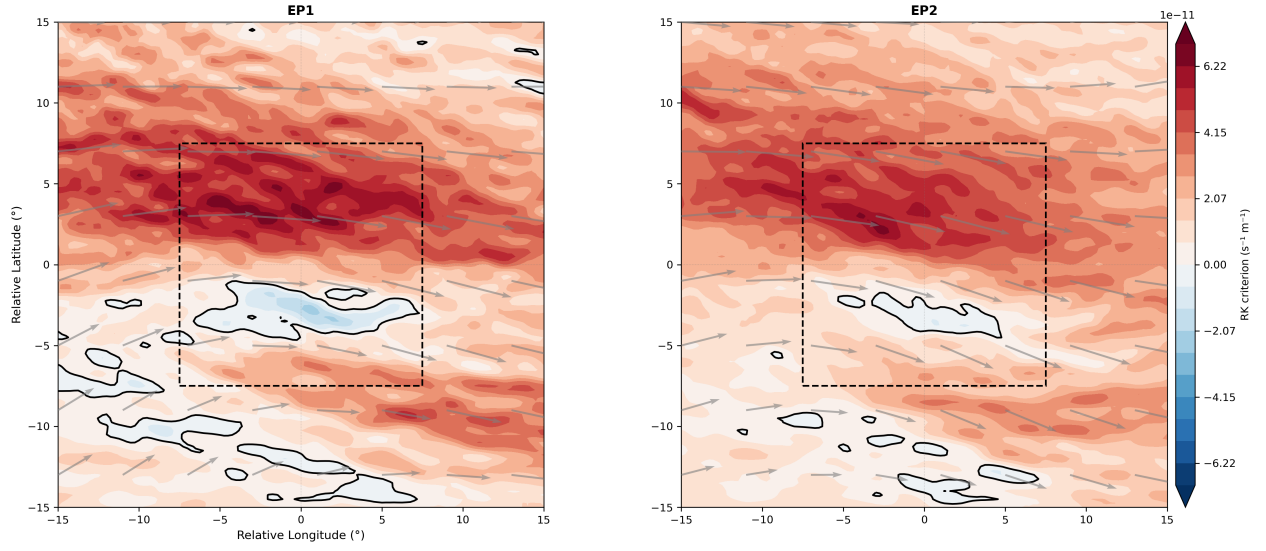


Figure 7: RK Criterion Composite

Kinetic Energy Advection at 250 hPa — EP1 vs EP2

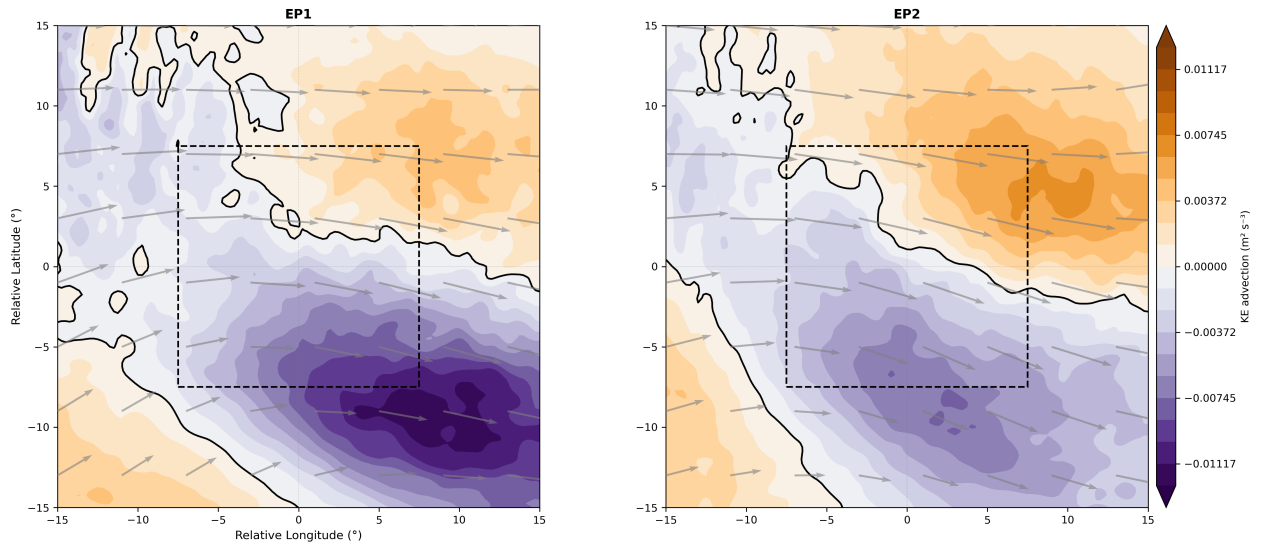


Figure 8: KE Advection Composite

Summary Statistics:

Statistic	EP1	EP2
Mean ($\text{m}^2 \text{ s}^{-3}$)	-1.760e-03	-4.142e-04
Range ($\text{m}^2 \text{ s}^{-3}$)	[-1.241e-02, 4.948e-03]	[-7.759e-03, 7.122e-03]
LEC $15 \times 15^\circ$ mean	-2.622e-03	-1.380e-03
Full $30 \times 30^\circ$ mean	-1.760e-03	-4.142e-04

Lateral boundaries (flux assessment):

Boundary	EP1	EP2
North ($+7.5^\circ$)	1.207e-03	3.303e-03
South (-7.5°)	-7.365e-03	-5.867e-03
East ($+7.5^\circ$)	-1.957e-03	1.570e-03
West (-7.5°)	-1.308e-03	-1.924e-03

5. Physical Interpretation

Status: Analysis of spatial patterns and physical mechanisms is ongoing. Preliminary observations include:

5.1 EP1 Characteristics

- {EP1_INTERPRETATION}

5.2 EP2 Characteristics

- {EP2_INTERPRETATION}

5.3 Key Structural Differences

- {KEY_DIFFERENCES}

Future Work: Detailed analysis of spatial patterns, vertical coupling mechanisms, and relationship between energetics (from cluster analysis) and structural characteristics.

6. Computational Implementation

6.1 Grid Spacing on Spherical Earth

Function: `compute_spherical_grid_spacing(lat_1d, lon_1d)`

Computes accurate grid spacing accounting for Earth's spherical geometry:

```
# Meridional spacing (constant per latitude band)
dy = R_EARTH * Δ # meters
```

```
# Zonal spacing (varies with latitude)
dx = R_EARTH * cos( ) * Δ # meters
```

Quality checks: - Verifies latitude is monotonic (increasing or decreasing) - Verifies longitude is monotonic - Raises `ValueError` if non-monotonic - Gradient sign automatically correct regardless of coordinate direction

6.2 Divergence on Spherical Coordinates

For small domains ($\sim 30^\circ$), the simplified Cartesian approximation is valid:

$$\nabla \cdot \vec{F} \approx \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}$$

where dx and dy account for spherical geometry (Section 6.1).

Full spherical formula (not required for 30° domain):

$$\nabla \cdot \vec{F} = \frac{1}{R \cos \phi} \frac{\partial F_\lambda}{\partial \lambda} + \frac{1}{R \cos \phi} \frac{\partial (F_\phi \cos \phi)}{\partial \phi}$$

Our implementation uses the simplified form with accurate dx , dy — appropriate for mesoscale domains.

6.3 Unit Tracking

All calculations use **MetPy units** for automatic dimensional analysis:

```
# Example: moisture flux
qu = (q * units('kg/kg')) * (u * units('m/s'))
# → qu.units = 'kg / kg * m / s' (automatically tracked)
```

```
# Convert divergence to g/kg/s
div_q_si = (dqu_dx + dqv_dy) * units('kg/kg/s')
div_q_gkg = (div_q_si * 1000 * units('g/kg')).magnitude
```

Benefits: - Prevents unit errors (e.g., mixing K and $^\circ\text{C}$) - Documents unit transformations
- Explicit conversions (no hardcoded magic numbers)

6.4 Numerical Methods

- **Spatial derivatives:** `numpy.gradient()` with 2nd-order centred finite differences
- **Vertical derivatives:** Centred FD over 3 levels (e.g., 175–200–225 hPa for PV@200)
- **Interpolation:** Linear interpolation to regular 0.25° grid via `xarray.interp()`

6.5 Quality Control Filters

Diagnostic	Filter	Rationale
EGR	$N^2 > 10^{-6} \text{ s}^{-2}$	Exclude statically unstable regions
EGR	$\ \phi\ > 5^\circ$	Avoid near-equatorial ($f \approx 0$)
EGR	$\sigma_{EGR} < 5 \text{ day}^{-1}$	Cap unrealistic growth rates
PV	Valid only where θ monotonic	Ensure isentropic sorting

6.6 Code Verification

Test cases: 1. Zonal wind $\rightarrow$ zero meridional derivative 2. Meridional wind $\rightarrow$ zero zonal derivative

3. Uniform field $\rightarrow$ zero divergence 4. Reversed latitude (90 $\rightarrow$ -90) $\rightarrow$ correct gradient sign

Validation: - Cross-check EGR values with literature (typical: 0.3–1.0 day⁻¹) - PV@200 should align with tropopause ($\sim$ 2–3 PVU contour) - Compare temperature advection with operational analyses

7. References

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